

SENSITIVITY-INFORMED MULTI-OBJECTIVE ADAPTIVE WEIGHTING METHOD FOR AUTOMATIC ALIGNMENT OF FREEFORM USTP SYSTEMS: SUPPLEMENTAL DOCUMENT

Figure S1 illustrates the functionally designed projection pattern and its corresponding image-processing workflow. Overall, the procedure follows the sequence of target-region extraction, coarse localization, fine localization, and MTF identification and calculation, so that both geometric-distortion evaluation and resolution evaluation can be carried out on the same projected image.

Figure S1(a) shows the original projection pattern, which includes the outer frame, the cross-shaped positioning structure, and the line-pair detection array. Figure S1(b) corresponds to the target-region extraction stage, where dynamic-threshold segmentation and region filling are mainly employed to obtain a stable region of interest (ROI). Figures S1(c) and S1(d) correspond to the coarse- and fine-localization stages, respectively. To improve localization accuracy, least-squares fitting is further adopted in the fine-localization stage to refine the corner points and boundary positions, thereby establishing a reliable geometric reference. The line-pair array shown in Fig. S1(e) is used for MTF evaluation. After the geometric localization is completed, the target line-pair region is further identified automatically, as illustrated in Fig. S1(f), and the MTF value at 93 lp/mm is then calculated from the grayscale contrast. In this way, both the distortion metrics and the MTF metric can be synchronously extracted from the same structured pattern under a unified coordinate reference, thereby providing a consistent image-processing basis for closed-loop alignment.

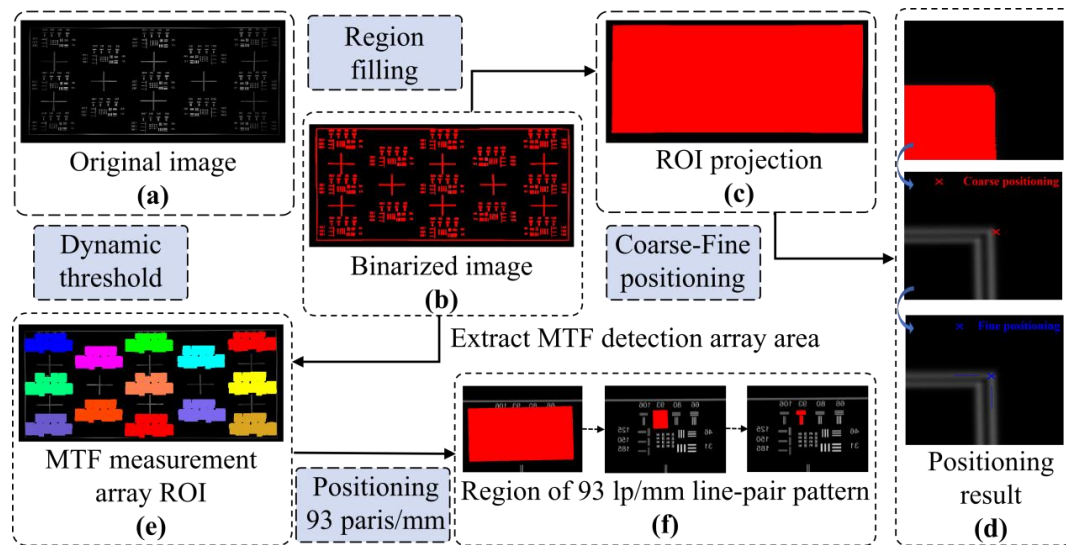


Fig. S1. Functionally designed projection pattern and the corresponding image-processing flow. (a) Original projection pattern. (b) Target-region extraction result after dynamic-threshold segmentation and region filling. (c) Coarse positioning result based on the positioning blocks. (d) Fine positioning result based on local line fitting and iterative intersection estimation. (e) MTF detection array used for the evaluation at 93 lp/mm. (f) Automatic recognition of the line-pair region and contrast analysis for MTF estimation.